

CULPEPER, VA, USA

WMO: 723098

Lat: 38.527N

Lon: 77.859W

Elev: 316

StdP: 14.53

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93798

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	17.7	-4.2	4.5	22.5	1.1	5.9	24.6	19.8	40.4	17.9	39.6	1.7	0	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	94.7	76.5	91.5	75.4	90.2	75.1	80.7	88.3	78.9	87.3	77.5	86.4	5.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	152.1	84.4	76.7	140.8	82.8	75.0	133.0	81.2	44.4	88.5	42.6	86.8	41.1	86.4	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.9	14.5	12.0	DB	4.3	97.9	4.7	3.3	0.9	100.3	-1.8	102.2	-4.5	104.1	-7.9	106.5
			WB	3.7	82.5	4.7	2.1	0.3	84.1	-2.4	85.3	-5.0	86.5	-8.4	88.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.1	35.3	38.3	46.5	57.0	65.9	73.6	77.6	76.3	69.6	58.1	47.4	38.9
	DBStd	16.86	10.35	9.57	9.48	8.69	7.37	5.47	4.59	5.03	6.34	8.38	8.46	8.55
	HDD50	1556	472	345	185	28	0	0	0	0	0	20	146	360
	HDD65	4219	922	749	577	269	83	5	0	0	28	248	529	809
	CDD50	4160	15	17	75	238	492	708	854	817	589	270	69	16
	CDD65	1347	0	0	2	29	110	263	389	352	167	33	2	0
	CDH74	13390	2	7	61	457	1167	2593	3921	3338	1463	351	28	2
	CDH80	5528	0	0	8	144	403	1060	1815	1471	519	105	3	0
Wind	WSAvg	3.5	4.1	4.4	5.0	4.6	3.3	2.9	2.5	2.4	2.7	3.1	3.5	3.4
Precipitation	PrecAvg	41.3	3.0	2.2	3.8	3.3	4.1	4.0	4.1	3.3	4.1	3.1	3.2	3.2
	PrecMax	56.4	6.6	6.5	9.0	5.0	8.2	8.4	7.1	7.3	10.2	8.1	6.2	5.7
	PrecMin	29.4	1.0	0.6	0.0	1.6	1.4	1.4	1.3	1.2	0.0	0.1	0.9	1.6
	PrecStd	6.2	1.6	1.2	2.0	1.1	1.9	1.8	1.6	1.4	2.6	2.0	1.6	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	73.2	80.9	90.2	91.3	96.6	99.1	97.5	93.4	88.5	78.4	70.4
		MCWB	61.2	58.6	62.4	67.5	72.4	76.9	78.5	76.4	74.4	72.4	64.4	59.8
	2%	DB	63.8	65.9	74.5	84.0	88.3	91.4	95.0	93.4	90.1	82.1	72.3	63.7
		MCWB	56.8	54.8	58.8	65.1	72.0	75.3	77.1	76.4	73.8	67.9	61.1	57.3
	5%	DB	57.4	60.8	69.8	80.0	84.3	89.8	91.8	90.8	85.7	77.4	66.6	57.8
		MCWB	49.5	50.0	56.9	62.9	70.3	75.2	76.4	75.9	72.3	66.1	58.6	51.2
	10%	DB	52.1	54.5	64.1	74.2	81.2	86.5	89.9	88.2	82.0	72.9	63.3	54.0
		MCWB	45.8	46.0	53.4	60.7	68.7	73.5	76.1	74.9	70.3	63.8	55.3	48.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.8	62.3	65.7	71.2	77.3	81.3	82.5	82.4	79.9	74.6	68.7	64.9
		MCDB	67.3	70.2	73.2	79.4	87.6	87.6	90.9	90.0	85.4	82.0	72.8	66.8
	2%	WB	58.7	56.9	62.2	68.2	74.8	78.9	80.8	80.2	77.1	72.2	64.5	59.1
		MCDB	63.5	63.8	70.7	79.2	83.3	87.9	89.1	87.9	84.5	79.2	69.0	63.9
	5%	WB	51.6	51.6	58.9	65.7	72.8	77.2	79.0	78.4	75.0	69.7	60.9	53.4
		MCDB	56.4	57.8	67.5	76.2	81.6	86.7	87.8	86.3	80.9	74.3	65.7	56.5
	10%	WB	46.2	46.9	55.0	62.9	70.6	75.3	77.6	76.9	73.2	66.0	56.7	48.1
		MCDB	50.5	53.6	62.8	71.1	78.2	83.5	86.7	84.6	79.2	71.2	61.5	52.6
Mean Daily Temperature Range	5% DB	MDBR	20.2	21.8	23.6	24.6	22.4	21.6	20.8	20.9	21.4	23.3	23.1	20.2
		MCDBR	28.5	30.6	31.4	32.5	27.9	25.7	24.6	24.2	26.2	28.8	29.6	26.3
		MCWBR	20.3	20.5	18.3	15.7	13.5	11.8	10.1	9.8	11.8	15.9	18.9	19.1
	5% WB	MCDBR	23.5	25.7	27.0	26.6	23.4	22.5	21.4	20.2	20.2	22.3	22.5	21.9
		MCWBR	19.0	19.4	18.1	14.5	12.5	11.4	10.5	10.1	10.9	13.1	17.4	18.6
Clear-Sky Solar Irradiance	taub	0.300	0.317	0.353	0.394	0.434	0.490	0.507	0.509	0.418	0.356	0.320	0.305	
	taud	2.499	2.432	2.390	2.299	2.211	2.094	2.072	2.044	2.281	2.452	2.510	2.520	
	Ebn at Noon	280	289	288	280	270	254	248	244	262	271	270	270	
	Edh at Noon	25	30	35	41	45	51	51	52	38	30	24	23	
All-Sky Solar Radiation	RadAvg	680	939	1249	1559	1748	1925	1866	1675	1379	1053	770	599	
	RadStd	59	105	80	117	170	137	101	82	166	118	62	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+230	+137

Nomenclature: See separate page